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University of Colorado **Boulder**

Introduction to AI-Assisted Coding: Prompt Engineering with LLMs in Google Colab

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Supported by NSF grant #2321008



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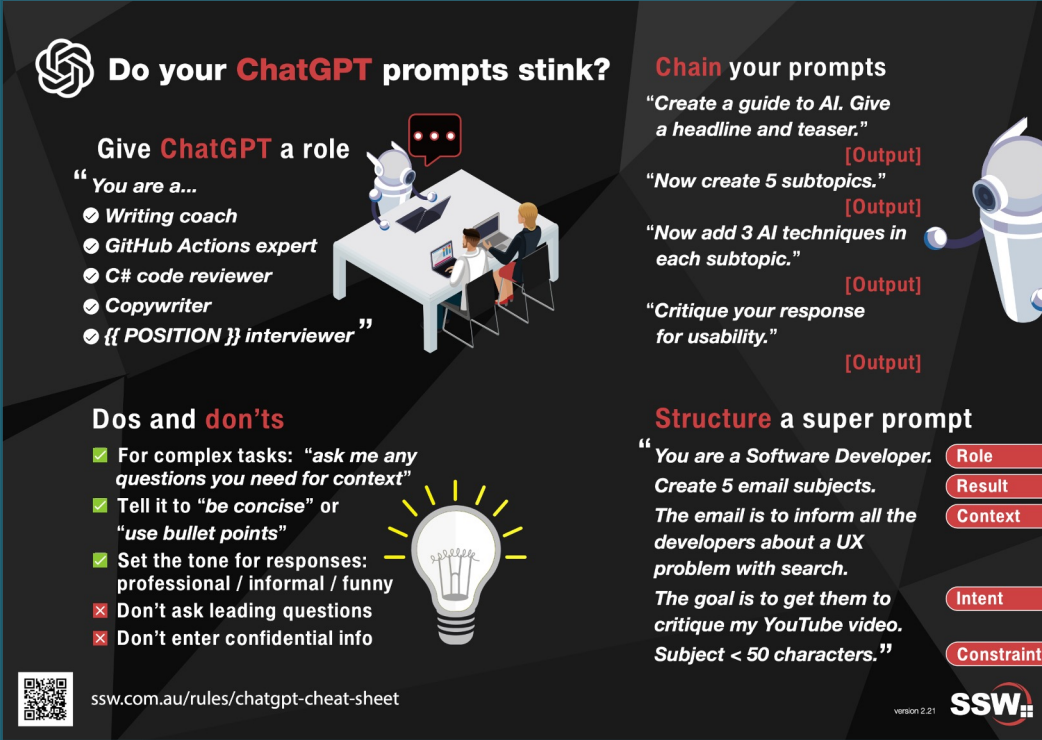
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Prompt Engineering Primer

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- **Prompt engineering** is the process of designing and optimizing the language input to effectively guide a LLM response
- Requires **Context** and **Expectations** and can include a **Persona** (role or perspective)
- The more context and detailed expectations you provide the better the output.



Do your ChatGPT prompts stink?

Give ChatGPT a role

"You are a..."

- Writing coach
- GitHub Actions expert
- C# code reviewer
- Copywriter
- {{ POSITION }} interviewer"

Dos and don'ts

- ✓ For complex tasks: "ask me any questions you need for context"
- ✓ Tell it to "be concise" or "use bullet points"
- ✓ Set the tone for responses: professional / informal / funny
- ✗ Don't ask leading questions
- ✗ Don't enter confidential info

Chain your prompts

"Create a guide to AI. Give a headline and teaser."

[Output]

"Now create 5 subtopics."

[Output]

"Now add 3 AI techniques in each subtopic."

[Output]

"Critique your response for usability."

[Output]

Structure a super prompt

"You are a Software Developer. Create 5 email subjects. The email is to inform all the developers about a UX problem with search. The goal is to get them to critique my YouTube video. Subject < 50 characters."

Role
Result
Context
Intent
Constraint

ssw.com.au/rules/chatgpt-cheat-sheet

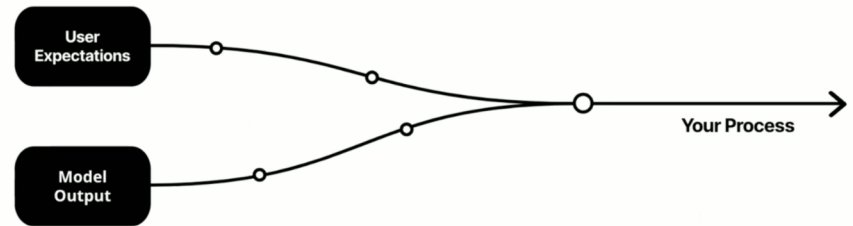
version 2.21

SSW

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- **Prompt engineering** is usually an iterative process
- Can benefit from:
 - **Delimiters** can highlight logical breaks and tasks
 - **Step-by-step instructions**
 - **One/few shot learning** (give it examples of what you want)
 - **Ask the LLM** to help write an effective prompt!

Iterative Approach



Powered by OpenAI

<https://academy.openai.com/public/videos/advanced-prompt-engineering-2025-02-13>

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Goals for the Tutorial



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Goals for the Tutorial

- **Have FUN!!**
- Use large language models (LLMs) to generate and modify Python code in Google Colab
- Apply prompt-engineering strategies to iteratively debug code, resolve errors, and improve analytical workflows
- Integrate satellite remote sensing data (NDVI from NASA's Harmonized Landsat–Sentinel dataset) with U.S. Census socioeconomic datasets
- Summarize and visualize spatial patterns using maps, tables, and simple statistical comparisons
- Develop skills for using AI to write and refine code while creating a reproducible workflow that can be adapted to other geospatial datasets and research questions.



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Visual Roadmap for the Tutorial



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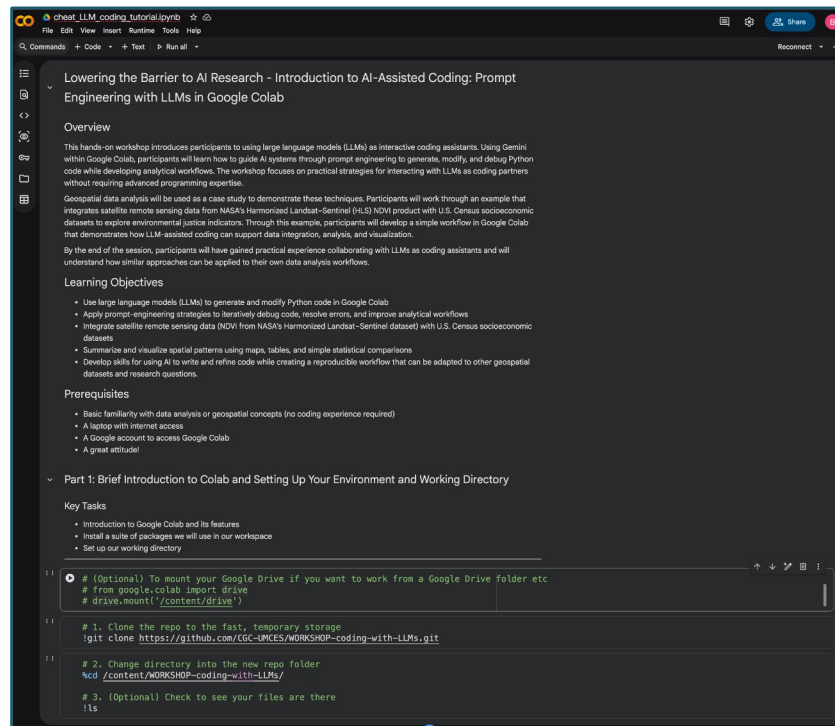
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Part 1: Introduction to Google Colab environment



Baltimore City and County Census Tracts we will use



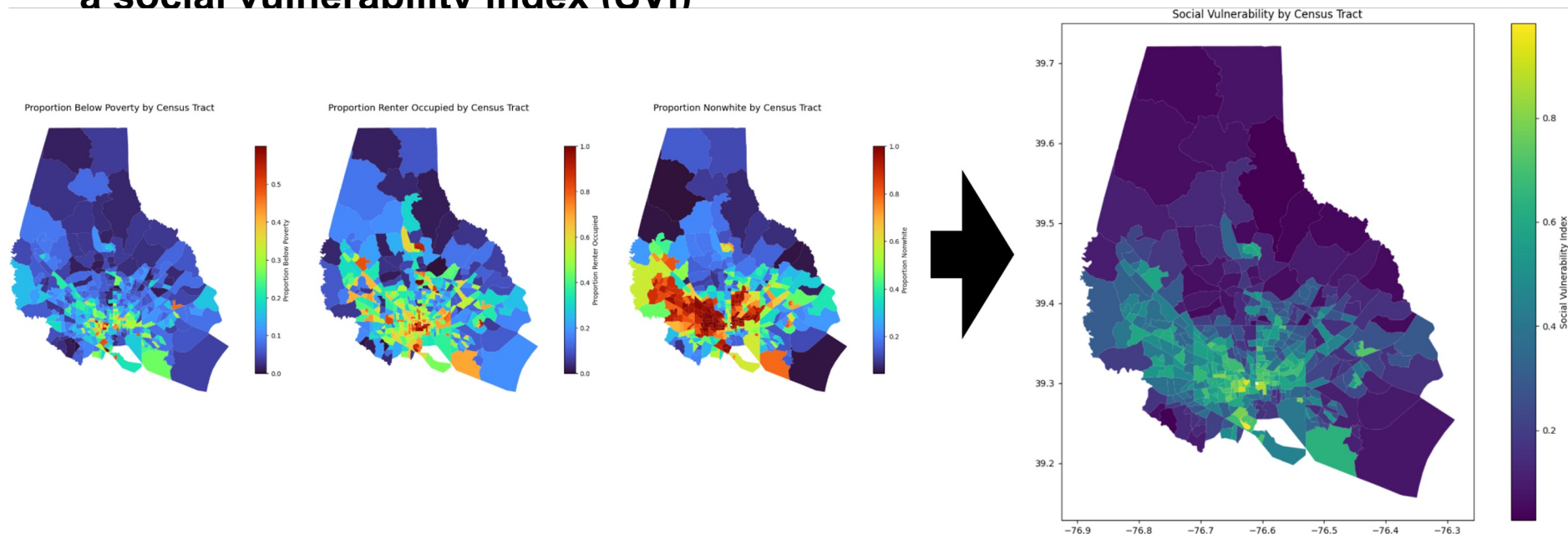
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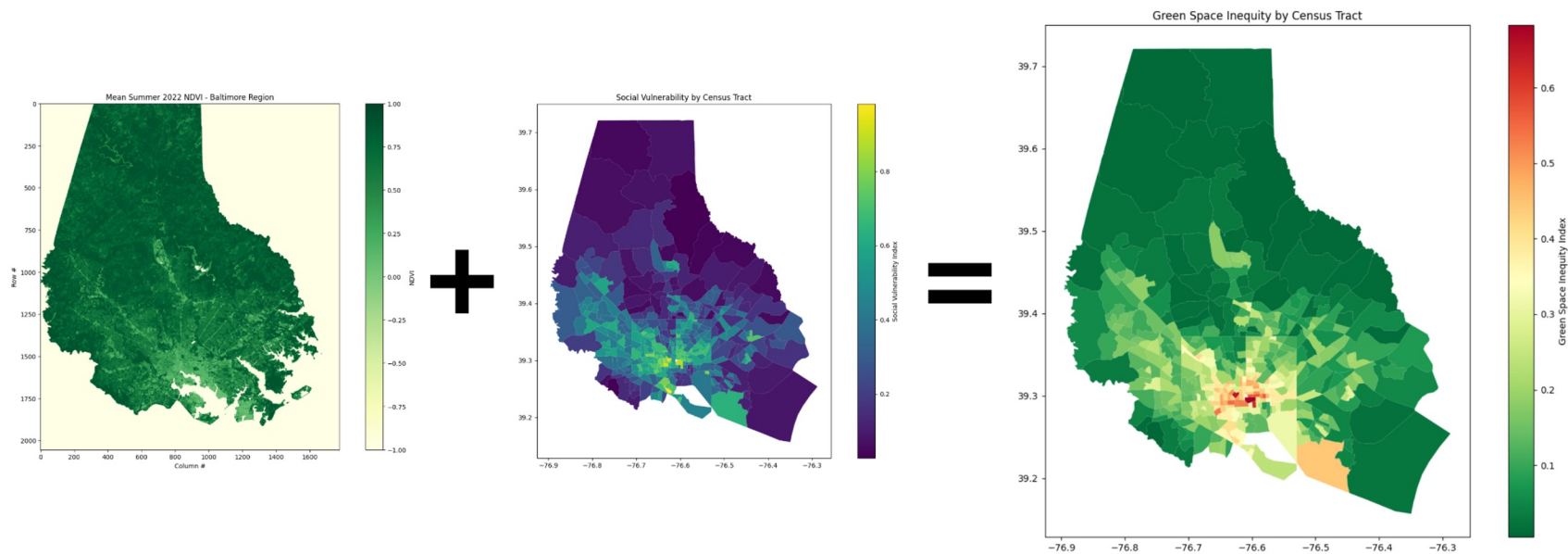
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Part 2: Working with Census data to create a social vulnerability index (SVI)



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Part 3: Integrating Harmonized Landsat-Sentinel 2 NDVI data with the SVI from Part 2 to build a Green Space Inequity Index



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Part 4: Freestyling with LLMs!



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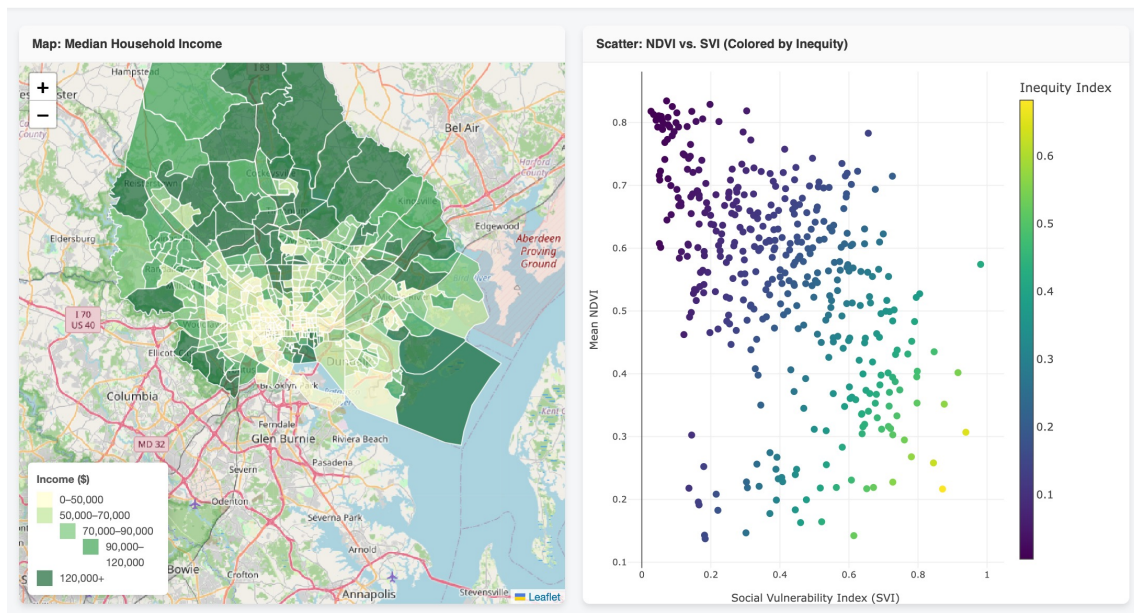
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Part 5: Creating a Shareable Interactive Data Dashboard

Interactive Green Space Inequity Analysis

Lasso points on the scatter plot to filter/highlight Census Tracts on the map.



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Discussion Q & A (last ~5-10 mins)

- Discussion of the strengths and limitations of LLM-assisted analysis
- Best practices for validating AI-generated code and maintaining reproducibility
- Opportunities and challenges for AI in geospatial research
- Audience questions and discussion

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Thank You!

Questions?



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Github Repository URL

<https://github.com/CGC-UMCES/NAIRR-2026-Tutorial-Intro-Colab-LLM-Coding>



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